Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

PremierLeague

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Does Stadium Capacity Relate to Club Performance? Premier League 2025/26

1. Introduction

We wanted to check a simple idea that fans often repeat: clubs with bigger stadiums do better. Our research question was whether the capacity of a club's home stadium is associated with its results in a single season. We took the 2025/26 Premier League (15 Aug 2025 – 24 May 2026, 38 matchdays, 20 clubs) and used the number of points in the final table as the measure of performance.

2. Data sources

We combined two sources of different types:

#	Source	Method	What we took
1	football-data.org API (v4), endpoint /competitions/PL/standings	requests + JSON	position, played, wins, draws, losses, goals for/against, points
2	Wikipedia, List of Premier League stadiums	requests + BeautifulSoup	club, stadium, capacity

Before scraping, we checked Wikipedia's robots.txt with urllib.robotparser, and it allowed access to the page. We also sent a descriptive User-Agent header.

3. Cleaning and merging

API data. We flattened the JSON with json_normalize, renamed the columns (for example team.name to club_api, won to wins) and kept only the ones we needed. There were no missing values and no duplicate clubs.

Scraped data. The Wikipedia table has several problems. First, it contains 63 rows, including closed historical stadiums. We kept only rows with an empty "closed" column, which left 52. Without this step, clubs could be matched to an old ground such as Highbury or the Boleyn Ground. Second, capacity is stored as text with commas and footnote markers (for example "61,276[1]"). We removed the markers and commas with regular expressions and converted the values to integers.

Merging. Club names differ between the sources ("Manchester United FC" vs. "Manchester United"). We built a normalized key for each club: lowercase, accents removed, "&" replaced by "and", and the tokens FC/AFC/CF/the removed. We used difflib fuzzy matching as a safety net for near misses. One name had to be fixed by hand: "Crystal Palace Wimbledon" on Wikipedia was mapped to Crystal Palace. We merged on the key with validate="one_to_one". All 20 clubs matched and no capacity was missing.

Manual correction. Wikipedia lists both of Everton's stadiums as active, and our duplicate filter kept the first one, Goodison Park. Everton played the 2025/26 season at Hill Dickinson Stadium, so we set its capacity to 52,769.

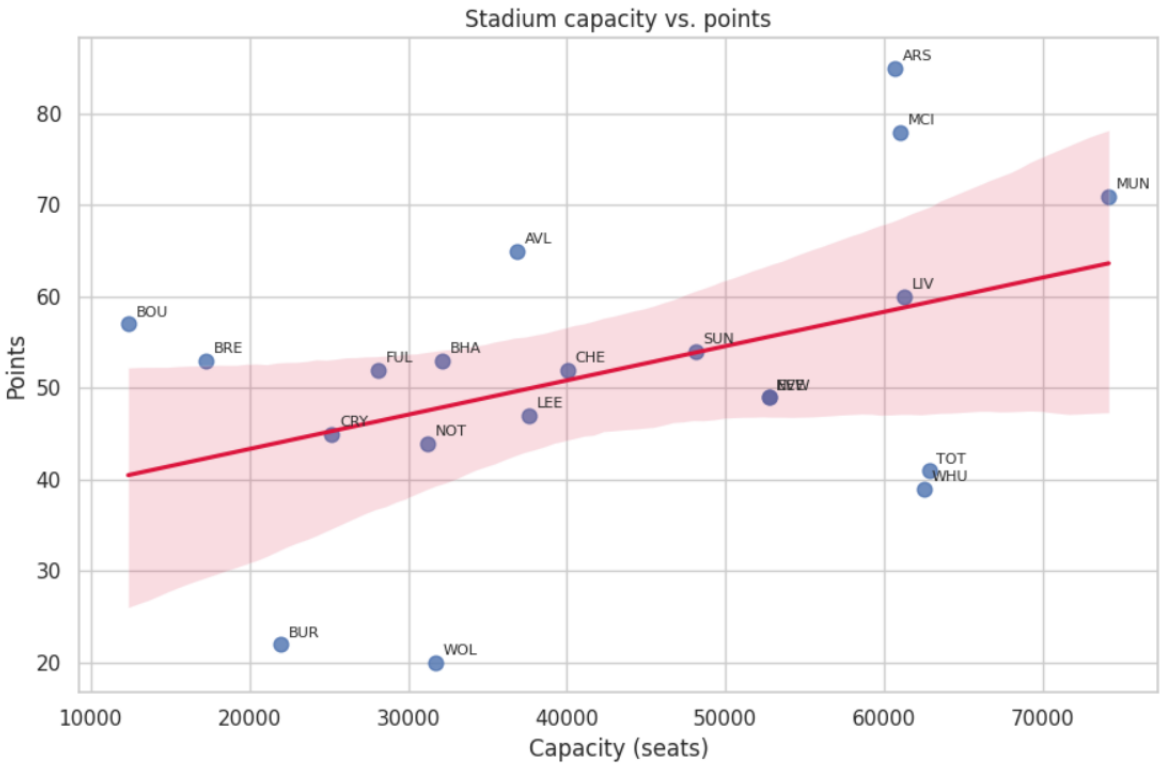
New features. We added points_per_game and size_group. The second splits the clubs into three groups (Small, Medium, Large) with pd.qcut on capacity. The final dataset is saved as clean_merged.csv.

position	playedGames	form	won	draw	lost	points	goalsFor	goalsAgainst	goalDifference	team.id	team.name	team.shortName	team.tla	team.cres
1	38	W,W,W,W,W	26	7	5	85	71	27	44	57	Arsenal FC	Arsenal	ARS	https://crests.footbal data.org/57.pn
2	38	L,D,W,W,D	23	9	6	78	77	35	42	65	Manchester City FC	Man City	MCI	https://crests.footbal data.org/65.pn
3	38	W,W,D,W,W	20	11	7	71	69	50	19	66	Manchester United FC	Man United	MUN	https://crests.footbal data.org/66.pn
4	38	W,W,D,L,L	19	8	11	65	56	49	7	58	Aston Villa FC	Aston Villa	AVL	https://crests.footbal data.org/58.pn
5	38	D,L,D,L,W	17	9	12	60	63	53	10	64	Liverpool FC	Liverpool	LIV	https://crests.footbal data.org/64.pn

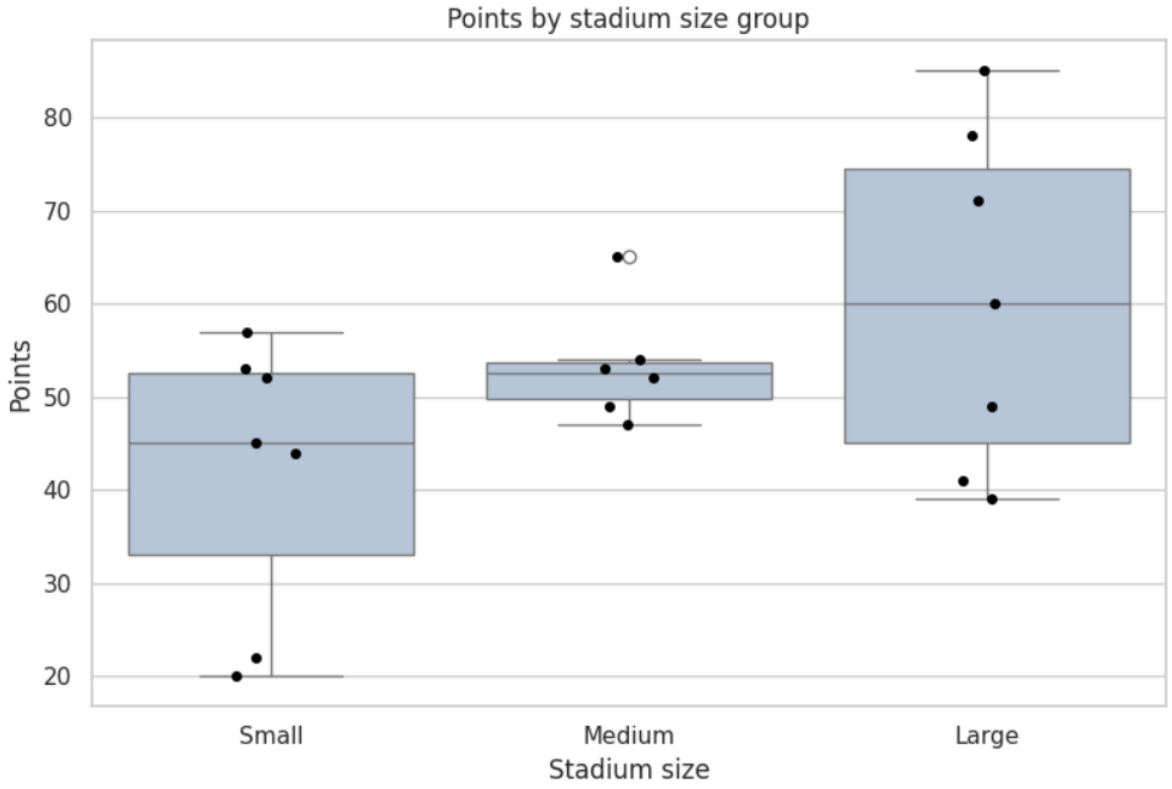
4. Results

Stadium capacity averages 42,536 seats (std 17,725), from 12,357 (Bournemouth) to 74,158 (Manchester United). Points average 51.8 (std 15.8), from 20 (Wolverhampton) to 85 (Arsenal).

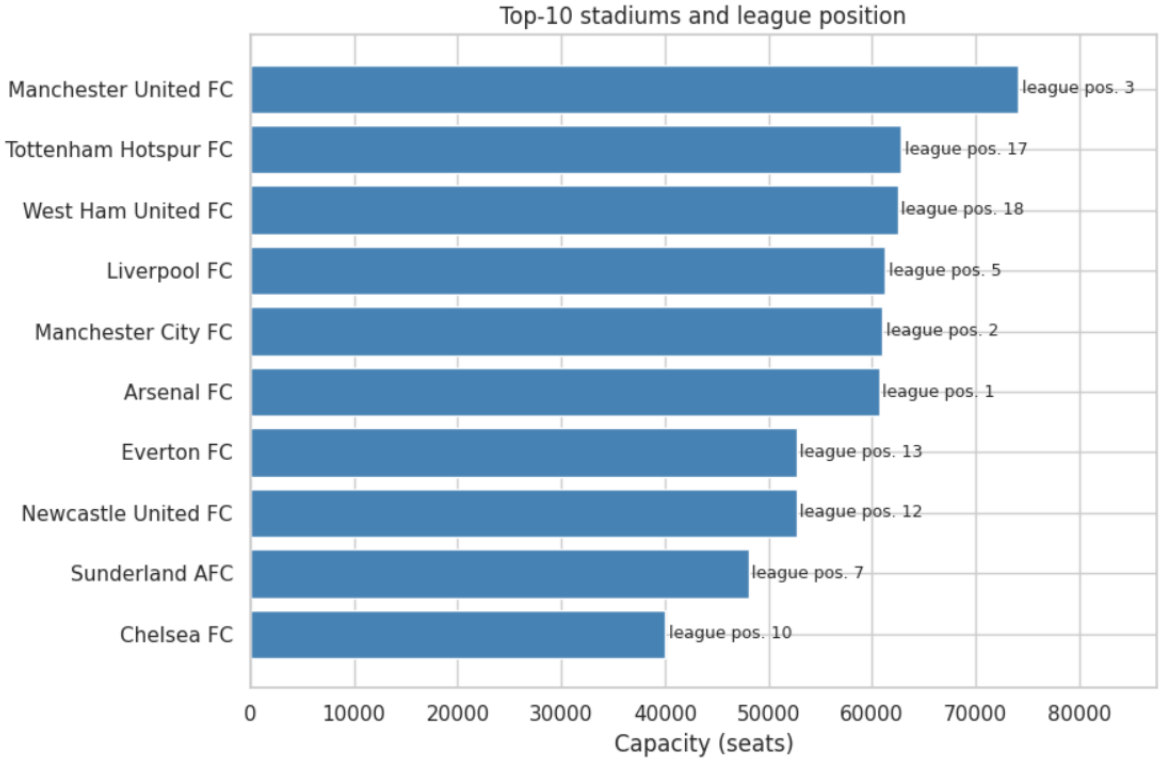
The correlation between capacity and points is positive but weak to moderate: Pearson r = 0.42 (p = 0.066), Spearman rho = 0.24 (p = 0.317). The linear trend is about +4 points per additional 10,000 seats, and capacity explains only around 18% of the variance in points ($r^2 \approx 0.18$). With 20 clubs this is not significant at the 5% level, and the rank-based Spearman coefficient is even weaker. Since every club played 38 games, points per game gives exactly the same correlations as points, so we use total points.



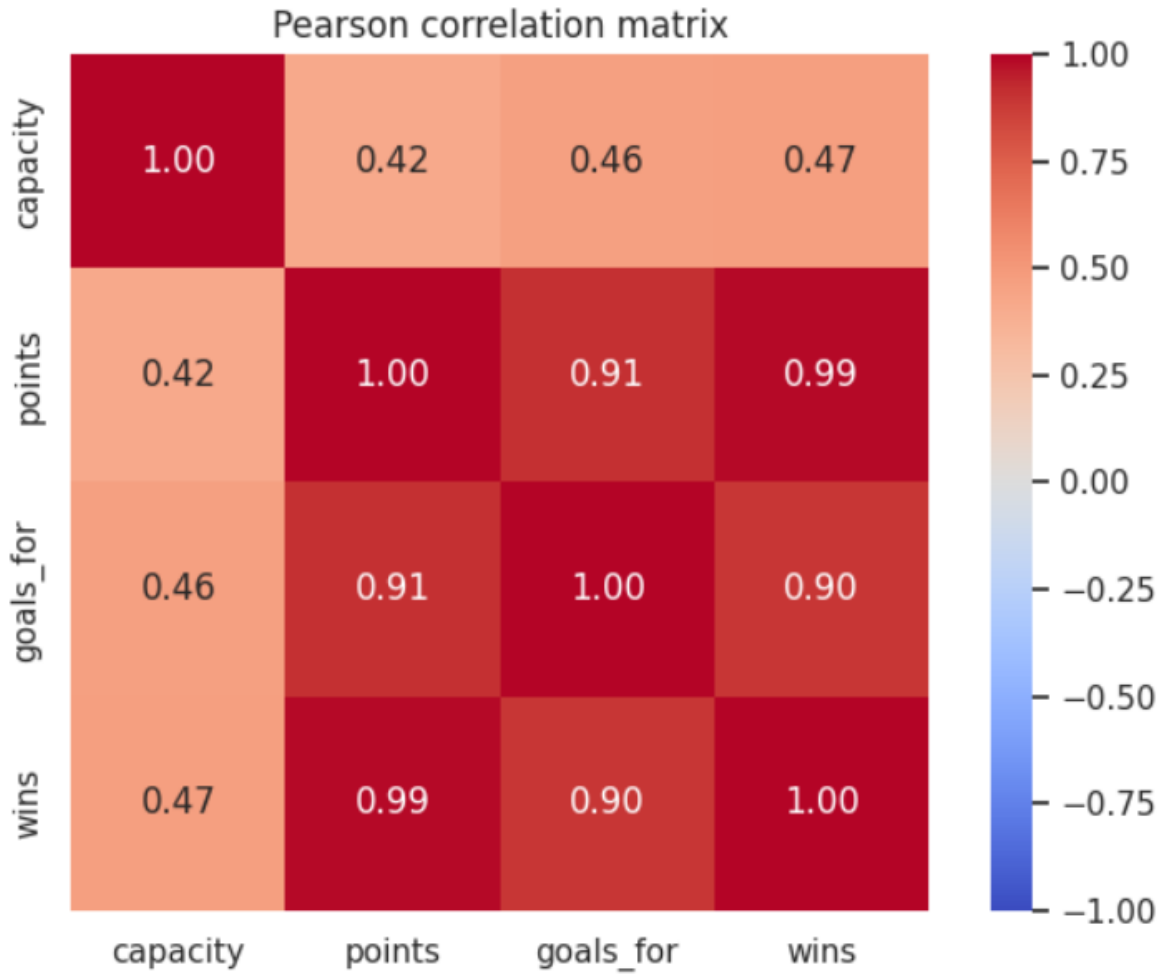
When we split the clubs by stadium size, median points grow from about 45 (small) to about 53 (medium) and 60 (large). The large group, however, is very spread out, from roughly 39 to 85 points, so a big stadium guarantees nothing.



To see which clubs break the pattern, we calculated standardized residuals from the regression line. Arsenal (60,704 seats, 85 points, z = +1.84), Manchester City (z = +1.34) and Bournemouth (the smallest stadium, 57 points, z = +1.15) perform better than their stadium would predict. Wolverhampton (z = -1.93), Burnley (z = -1.54) and West Ham (62,500 seats, only 39 points, z = -1.41) perform worse. Tottenham also has a very large stadium (62,850) and finished 17th.



Finally, the correlation matrix shows that capacity correlates 0.42 with points, 0.46 with goals scored and 0.47 with wins. Points are almost identical to wins (0.99) and strongly tied to goals scored (0.91).



5. Conclusions and limitations

In 2025/26, stadium capacity is positively but only weakly related to results, and the relationship is not statistically significant. We do not claim that a bigger stadium produces better results: correlation is not causation. Big stadiums usually belong to large, wealthy clubs, and money (players, coaches) is a more likely explanation. Club budget is the hidden factor we did not measure.

Limitations: only one season and 20 clubs; capacity is the current capacity, not real attendance; Wikipedia data can be outdated, which is why one value (Everton) had to be corrected by hand. For future work we would add other leagues (La Liga, Bundesliga, Serie A) and several seasons, and control for wage bill or revenue.

6. Team contribution

Nurbakyt : API part.
Nurassyl : web scraping.
Yerbol : cleaning and merging.
Nurdaulet : analysis, visualization and report.